

Vegan proteins may reduce risk of cancer, obesity, and cardiovascular disease by promoting increased glucagon activity.

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Amino acids modulate the secretion of both insulin and glucagon; the composition of dietary protein therefore has the potential to influence the balance of glucagon and insulin activity. Soy protein, as well as many other vegan proteins, are higher in non-essential amino acids than most animal-derived food proteins, and as a result should preferentially favor glucagon production. Acting on hepatocytes, glucagon promotes (and insulin inhibits) cAMP-dependent mechanisms that down-regulate lipogenic enzymes and cholesterol synthesis, while up-regulating hepatic LDL receptors and production of the IGF-I antagonist IGFBP-1. The insulin-sensitizing properties of many vegan diets--high in fiber, low in saturated fat--should amplify these effects by down-regulating insulin secretion. Additionally, the relatively low essential amino acid content of some vegan diets may decrease hepatic IGF-I synthesis. Thus, diets featuring vegan proteins can be expected to lower elevated serum lipid levels, promote weight loss, and decrease circulating IGF-I activity. The latter effect should impede cancer induction (as is seen in animal studies with soy protein), lessen neutrophil-mediated inflammatory damage, and slow growth and maturation in children. In fact, vegans tend to have low serum lipids, lean physiques, shorter stature, later puberty, and decreased risk for certain prominent 'Western' cancers; a vegan diet has documented clinical efficacy in rheumatoid arthritis. Low-fat vegan diets may be especially protective in regard to cancers linked to insulin resistance--namely, breast and colon cancer--as well as prostate cancer; conversely, the high IGF-I activity associated with heavy ingestion of animal products may be largely responsible for the epidemic of 'Western' cancers in wealthy societies. Increased phytochemical intake is also likely to contribute to the reduction of cancer risk in vegans. Regression of coronary stenoses has been documented during low-fat vegan diets coupled with exercise training; such regimens also tend to markedly improve diabetic control and lower elevated blood pressure. Risk of many other degenerative disorders may be decreased in vegans, although reduced growth factor activity may be responsible for an increased risk of hemorrhagic stroke. By altering the glucagon/insulin balance, it is conceivable that supplemental intakes of key non-essential amino acids could enable omnivores to enjoy some of the health advantages of a vegan diet. An unnecessarily high intake of essential amino acids--either in the absolute sense or relative to total dietary protein--may prove to be as grave a risk factor for 'Western' degenerative

diseases as is excessive fat intake.